Reproducibility Report for Expectation-driven sensory adaptations support enhanced acuity during categorical perception by Tim Sainburg et. al. (2025, Nature Neuroscience)

Will Hammond (whammond@ucsd.edu)

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Introduction

Tim Sainburg's 2025 paper includes a breadth of modern techniques and analysis used in the Gentner Lab, making it a useful paper to simulate in order to familiarize myself with relevant concepts and data. I will be simulating behavioral data from a two-alternative choice task (2AC) where the choices are modulated by the reliabilty of an auditory primer cue. My main

analysis goal is to find a significant difference in the population psychometric curve inflection points for decision making based on the reliability of the primer cue. This simulation will engage my interest in sensory systems while allowing me to work with real data and get experience analyzing it.

For this project, I will need to simulate a new data set, clean it to calculate the necessary analytic points (such as change in inflection point), create a linear mixed effects model, and create a 2 distribution histogram (figure 3e in the original paper). I will be aiming to replicate analysis with statistics similar to those reported in the paper ($= -11.12$, s.e.m. $= 1.158$, $z = -9.6$, $P < 0.001$) I believe that this will be an adequate challenge because it is a large dataset, and I do not have any experience with linear mixed effects models. I will need to ensure that I am cleaning and analyzing such a large dataset accurately and take the necessary steps to understand linear mixed effects models. Another challenge will be choosing simulation parameters that accuratley model the behavioral data. Although the data is not available in this paper, it is available in another paper (Gorman et al 2026) that analyzes the same dataset, making it easier to simulate an accurate representation of the data.

Learning Goals: My main goal is to re-familiarize myself with data analysis in R. I have used the language a few times for smaller projects, but without consistent use, I stil do not feel confident in my programming abilities. Although I anticipate challenges due to my inexperience, my conceptual understanding of the data, paper, and general programming knowledge make the scope of this project feasible for me. I also want to use this project as an opportunity to gain a greater understanding of linear mixed effects models, what they can tell us, and how to use them.

Links

Project repository: https://github.com/willhammond03/sainburg2026.git

Original paper: https://doi.org/10.1038/s41593-025-01899-1

Github repository for paper with accessible data: https://github.com/juliagorman/anticipatory_geometry_paper.git

Methods

Description of the steps required to reproduce the results

To reproduce the key result from this study, I will first need to change the data into a usable form. The pickle files need to be converted into files that R can use and stored in the repository instead of a local machine. I will need to create a dataframe from these individual data files, and this dataframe will benefit from being cleaned to remove extraneous variables. I will need to keep the “prob cue,” “correct,” and other values that I will need to discuss with the author to

reduce ambiguity. After cleaning, I will need to simulate new data based on the parameters of the real data. I will keep experimental values the same, and only change response values based on parameters observed in the original data. After simulating, I will make a new dataframe for the the dual histogram from figure 3e. This will only use values from probability 0.875 and 0.75 and their corresponding response. This figure is strictly for interpretation purposes and does not have associated statistical tests. Axis in the paper are imprecise, so confirmation of replication will need to be done visually. Next, I will make a dataframe to recreate figure 3f, including probability of accurate cue and inflection value. This visualization includes individual trial data, averaged individual data, averaged population data, and standard population level error. I will need to design a pipeline to include these statistics. I will need to fit a trendline using a linear mixed effects model. The model tests psychometric inflection point changing as a result of the main effect of cue probability while accounting for random effects of subject and cue response per subject. I plan on calculating the same statistics from the linear mixed effects model that are reported in the paper: cue probability: $= -11.12$, s.e.m. $= 1.158$, $z = -9.6$, $P < 0.001$.

Differences from original study

The original study runs these analyses and visualizations in Python, meaning that I will need to rebuild the code for it to fit into my analysis in R. I anticipate that this difference may change the appearance and aesthetics of my data visualizations, but I do not anticipate it significantly changing the statistical values returned. Running a simulation introduces structured variation, so I anticipate values in my figures and statistics to look slightly different, but not to an extent that alters the findings to a significant degree. The large effect size and the small p-value lead me to believe that simulating the data will not significantly change the statistical returns.

Project Progress Check 1

Measure of success

Please describe the outcome measure for the success or failure of your reproduction and how this outcome will be computed.

Pipeline progress

Earlier in this report, you described the steps necessary to reproduce the key result(s) of this study. Please describe your progress on each of these steps (e.g., data preprocessing, model fitting, model evaluation).

Results

Data preparation

Data preparation following the analysis plan.

Key analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Reproduction Attempt

Open the discussion section with a paragraph summarizing the primary result from the key analysis and assess whether you successfully reproduced it, partially reproduced it, or failed to reproduce it.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis of the dataset, (b) assessment of the meaning of the successful or unsuccessful reproducibility attempt - e.g., for a failure to reproduce the original findings, are the differences between original and present analyses ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the reproducibility attempt (if you contacted them). None of these need to be long.